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R's Book

Ranford Skyline's Book Template

Fully customised document template including:

- Custom document hierarchy: book, part, chapter, section, subsection.
- Custom tables of contents.
- Custom fonts.
- Support for:
  - Linguistics,
  - Hyperlinks,
- Some other stuff.

Table of Contents

I. **Examples** 3
- 1. Linguistics 4
- 2. Music 5
- 3. Code 6
- 4. Fonts 7
- 5. Maths 8

II. **Commands** 9

III. **Scatter** 10
- 1. Testing 13

Volume I

Examples

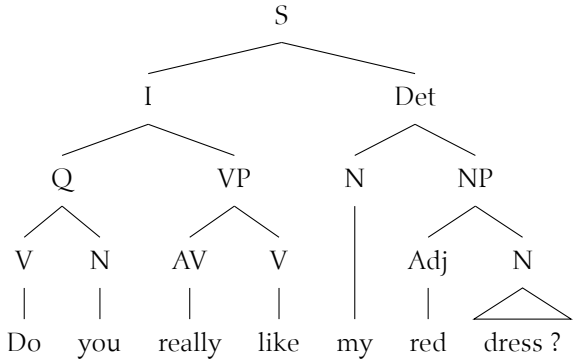
Table of Contents

1. **Linguistics** 4
2. **Music** 5
3. **Code** 6
4. **Fonts** 7
5. **Maths** 8

Chapter 1

Linguistics

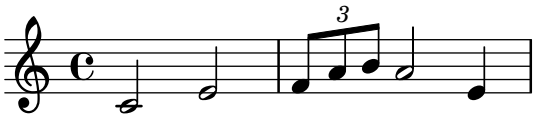
Phonetic transcriptions should use the `\ipa` command, i.e. `/ɹʌka/`. Use the forest environment with the linguistics default and the syntax setting to create syntax trees, such as:



Chapter 2

Music

All music transcriptions should be written in their own file, and included using the `\Rmusic` command. Music transcriptions are processed by lilypond.



Chapter 3

Code

Code imports handled by minted.

```
return function (path)
    local file = io.open (path, "r")
    if file then
        tex.sprint ("\\input{".. path .."}")
    else
        tex.sprint ("\\TODO{".. path .." unavailable}")
    end
end
```

Chapter 4

Fonts

Standard

ABCDEFGHIJKLMNOPQRSTUVWXYZ
abcdefghijklmnopqrstuvwxyz
0123456789

Small Caps

ABCDEFGHIJKLMNOPQRSTUVWXYZ
ABCDEFGHIJKLMNOPQRSTUVWXYZ
0123456789

Italics

ABCDEFGHIJKLMNOPQRSTUVWXYZ
abcdefghijklmnopqrstuvwxyz
0123456789

Math Mode

ABCDEFGHIJKLMNOPQRSTUVWXYZ
abcdefghijklmnopqrstuvwxyz
0123456789

Bold

ABCDEFGHIJKLMNOPQRSTUVWXYZ
abcdefghijklmnopqrstuvwxyz
0123456789

Math Fraktur

𝔄𝔅ℭ𝔇𝔈𝔉𝔊ℌℑ𝔍𝔎𝔏𝔐𝔑𝔒𝔓𝔔ℜ𝔖𝔗𝔘𝔙𝔚𝔛𝔜ℨ
𝔞𝔟𝔠𝔡𝔢𝔣𝔤𝔥𝔦𝔧𝔨𝔩𝔪𝔫𝔬𝔭𝔮𝔯𝔰𝔱𝔲𝔳𝔴𝔵𝔶𝔷
0123456789

Bold Italics

ABCDEFGHIJKLMNOPQRSTUVWXYZ
abcdefghijklmnopqrstuvwxyz
0123456789

Math Calligraphic

𝒜ℬ𝒞𝒟ℰℱ𝒢ℋℐ𝒥𝒦ℒℳ𝒩𝒪𝒫𝒬ℛ𝒮𝒯𝒰𝒱𝒲𝒳𝒴𝒵

Math Backboard

𝔸𝔹ℂ𝔻𝔼𝔽𝔾ℍ𝕀𝕁𝕂𝕃𝕄ℕ𝕆ℙℚℝ𝕊𝕋𝕌𝕍𝕎𝕏𝕐ℤ

Typewritter

ABCDEFGHIJKLMNOPQRSTUVWXYZ
abcdefghijklmnopqrstuvwxyz
0123456789

IPA

ABCDEFGHIJKLMNOPQRSTUVWXYZ
abcdefghijklmnopqrstuvwxyz
0123456789

Chapter 5

Maths

Denote vectors, direction vectors, and matrices with `\Rvec`, `\Rdir`, and `\Rmat`.

$$\underline{x} = \underline{\underline{M}} \cdot \underline{\underline{ab}}.$$

Differentiation with `Rdiff`, `\Rpdiff`, and `\RDiff`

$$\frac{dy}{dx} = \frac{d^2y}{dx^2}$$
$$\frac{\partial y}{\partial x} = \frac{\partial^2 y}{\partial x^2}$$
$$\frac{Dy}{Dx} = \frac{D^2y}{Dx^2}$$

Vec subscript test

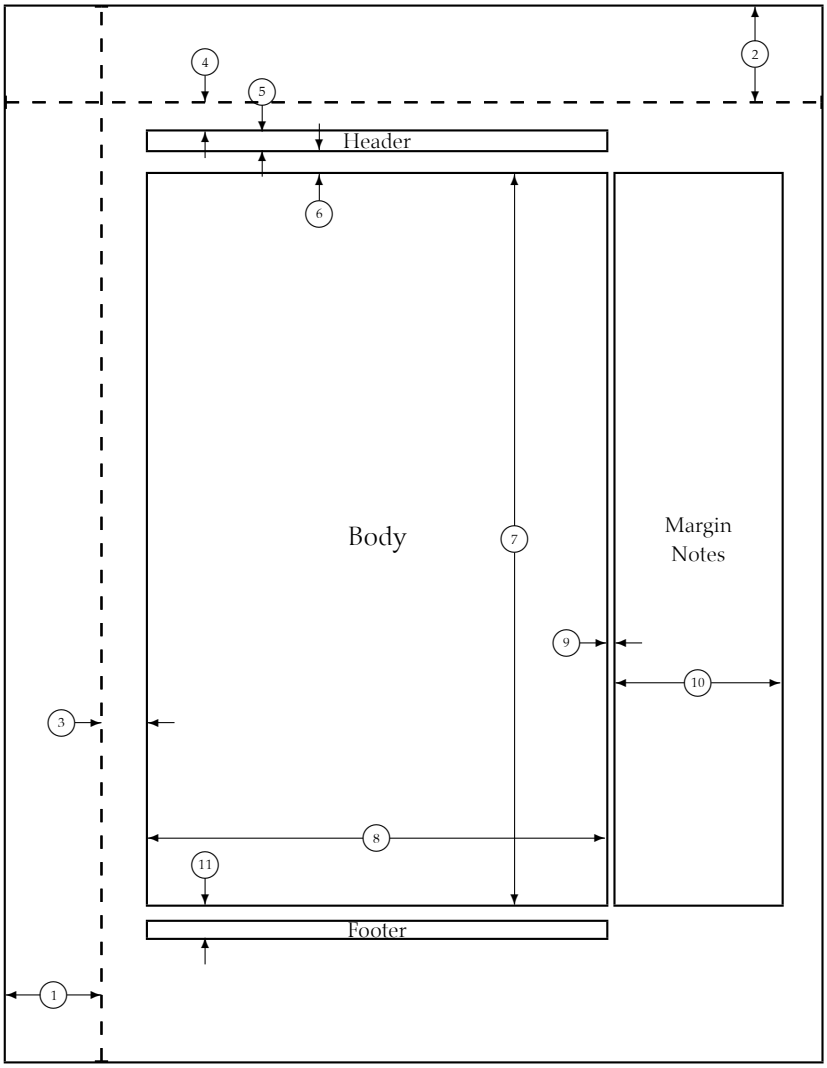
$$\underline{p}_{\underline{p}} \underline{\underline{A}}_{\underline{\underline{A}}_{\underline{\underline{A}}}}$$

Volume II

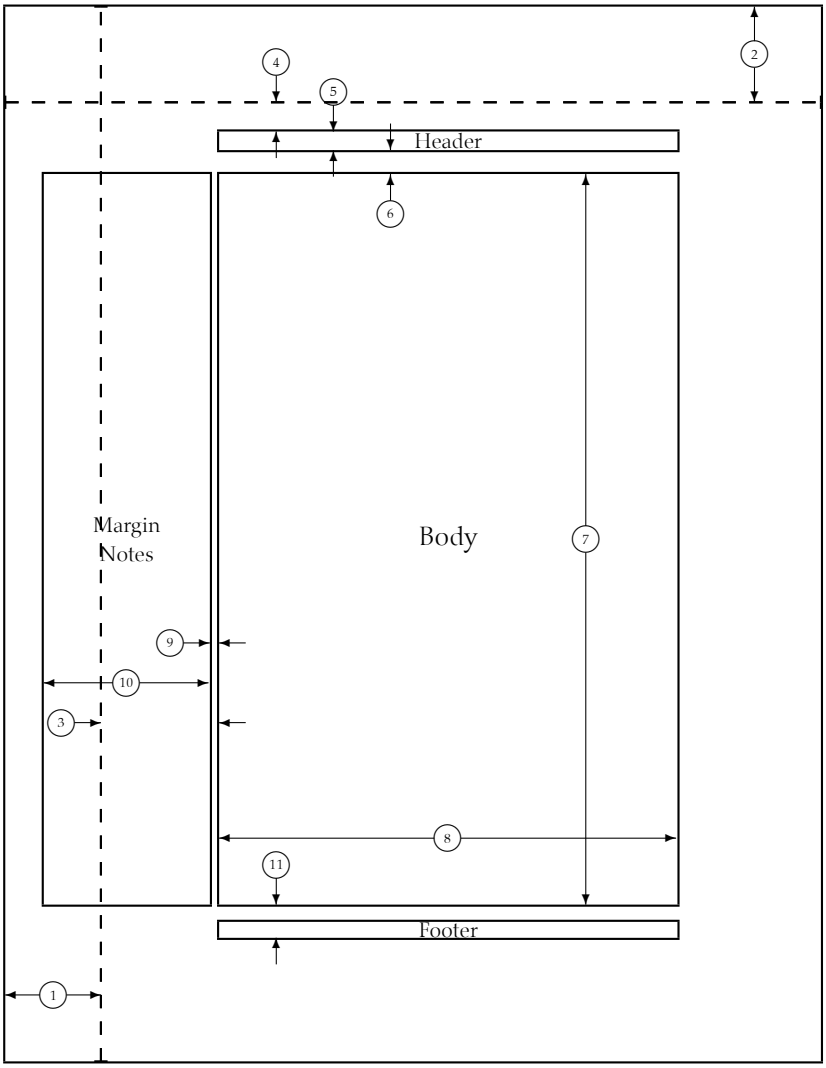
Commands

Volume III

Scatter



1	one inch + \hoffset	2	one inch + \voffset
3	\oddsidemargin = 35pt	4	\topmargin = 22pt
5	\headheight = 14pt	6	\headsep = 18pt
7	\textheight = 550pt	8	\textwidth = 345pt
9	\marginparsep = 7pt	10	\marginparwidth = 125pt
11	\footskip = 25pt		\marginparpush = 5pt (not shown)
	\hoffset = 0pt		\voffset = 0pt
	\paperwidth = 614pt		\paperheight = 794pt



1	one inch + \hoffset	2	one inch + \voffset
3	\evensidemargin = 89pt	4	\topmargin = 22pt
5	\headheight = 14pt	6	\headsep = 18pt
7	\textheight = 550pt	8	\textwidth = 345pt
9	\marginparsep = 7pt	10	\marginparwidth = 125pt
11	\footskip = 25pt		\marginparpush = 5pt (not shown)
	\hoffset = 0pt		\voffset = 0pt
	\paperwidth = 614pt		\paperheight = 794pt

Chapter 1

Testing

Hello World!

$$B_n(t) = \sum_{i=0}^{n} \binom{n}{i} t^i \left(\sum_{j=0}^{i} \binom{i}{j} x_j (-1)^{i-j} \right)$$

$$\begin{aligned} 0 &= \partial_\mu \left| \underline{l}(\mu) - \underline{p} \right| \\ \left| \underline{l} - \underline{p} \right|^2 &= \left| \underline{a} + \mu \underline{\underset{\smile}{ab}} - \underline{p} \right|^2 \end{aligned}$$

Quaternions are defined by the equation

$$\imath^2 = \jmath^2 = \kappa^2 = \imath\jmath\kappa = -1.$$

By considering pairs of terms, it is found that:

$$\begin{aligned} \imath\jmath\kappa &= \imath^2, & \jmath\kappa &= \imath, \\ \imath^2 = \jmath\kappa\imath &= \jmath^2, & \kappa\imath &= \jmath, \\ \imath\jmath\kappa &= \kappa^2, & \imath\jmath &= \kappa. \end{aligned}$$

By multiplying pairs of terms, it is found that:

$$\begin{aligned} \imath\jmath^2 &= \imath\jmath\jmath = \kappa\jmath = -\imath, \\ \imath^2\jmath &= \imath\imath\jmath = \imath\kappa = -\jmath, \\ \jmath^2\kappa &= \jmath\jmath\kappa = \jmath\imath = -\kappa. \end{aligned}$$

Note that the multiplication of quaternions is associative, but not commutative.

Kiŋ of þe dirt.

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Nulla malesuada porttitor diam. Donec felis erat, congue non, volutpat at, tincidunt tristique, libero. Vivamus viverra fermentum felis. Donec nonummy pellentesque ante. Phasellus adipiscing semper elit. Proin fermentum massa ac quam. Sed diam turpis, molestie vitae, placerat a, molestie nec, leo. Maecenas lacinia. Nam ipsum ligula, eleifend at, accumsan nec, suscipit a, ipsum. Morbi blandit ligula feugiat magna. Nunc eleifend consequat lorem. Sed lacinia nulla vitae enim. Pellentesque tincidunt purus vel magna. Integer non enim. Praesent euismod nunc eu purus. Donec bibendum quam in tellus. Nullam cursus pulvinar lectus. Donec et mi. Nam vulputate metus eu enim. Vestibulum pellentesque felis eu massa.

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